



UTM

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SAP BUSINESS TECHNOLOGY PLATFORM (BTP)

SECJ3744 – ENTERPRISE SYSTEMS DESIGN AND MODELING (WBL)

SECTION 01

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Introduction

SAP Business Technology Platform (BTP) is SAP's cloud-based Platform as a Service (PaaS) to integrate, automate, extend, and build AI-supported business applications and processes across enterprises. In this report, the detailed explanation of how SAP BTP is used to connect non-SAP applications using SAP Integration Suite. This is one of the tutorials provided by SAP BTP itself.

SAP Integration Suite is an Enterprise-grade integration Platform as a Service (EiPaaS) that allows to smoothly integrate on-premise and cloud-based applications and processes with tools and prebuilt content managed by SAP. SAP Integration Suite combines the integration capabilities such as Cloud Integration, API Management, Integration Advisor, Trading Partner Management, Integration Assessment, Open Connectors, and Migration Assessment into a cohesive and simplified toolkit for enterprise integrations. Thus, SAP Integration Suite is used in this topic of discussion.

Overview of The SAP BTP Tutorial

Figure 1 shows the use case scenario provided by the tutorial to depict the integration flow of reading information from a third-party software. Cloud Integration and Open Connectors which are part of SAP Integration Suite are used in this tutorial. Open Connectors are used to identify and establish connection to a third-party software and fetch data from the third-party software via APIs whereas Cloud Integration are used to design a custom integration flow to fetch data from Open Connectors and deploy the integration flow to a cloud-based runtime location. The integration flow then can be verified via the API by running the program. The API used in this tutorial is Postman. Figure 1 shows BambooHR being used as the third-party software. However, due to the failure of creating an account under BambooHR, Zoho CRM V2 was used instead as the third-party software to fetch data.

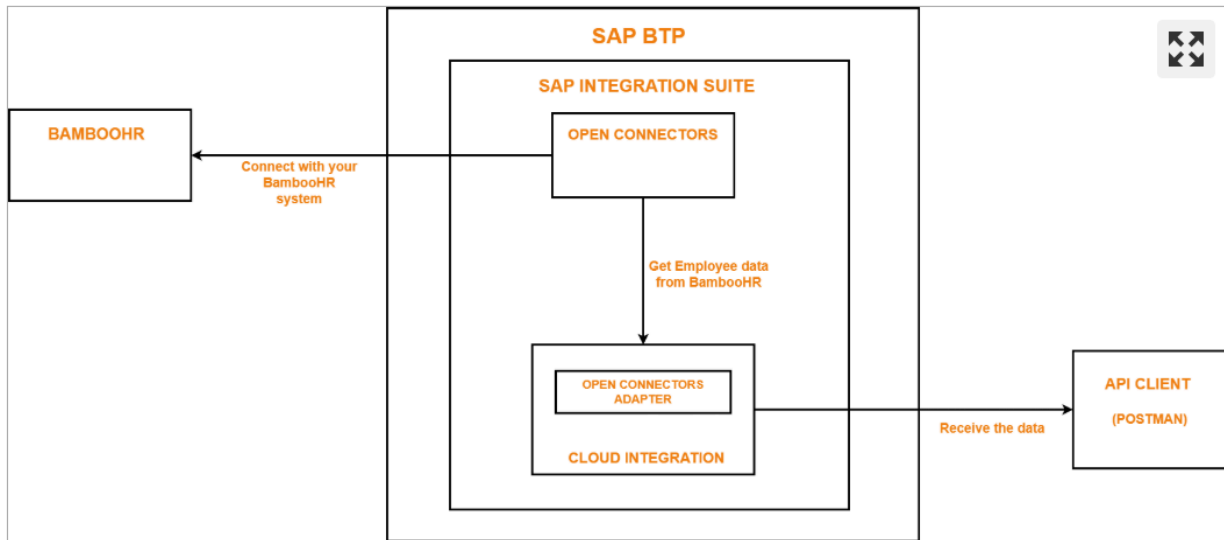


Figure 1 : Use Case Diagram of Connectivity to Non-SAP Applications using SAP Integration Suite

Step-by-step Explanation

1. Firstly, subscribe to SAP Integration Suite. Figure 2 shows the successful subscription of SAP Integration Suite.

The screenshot displays the SAP BTP Cockpit interface, specifically the 'Subaccount: trial - Instances and Subscriptions' page. The page shows a list of subscriptions, including 'Integration Suite' and 'SAP Business Application Studio', both with a status of 'Subscribed'. The 'Instances (0)' section below shows no instances created in the Cloud Foundry or Kyma/Kubernetes environments.

Application	Plan	Changed On	Status
Integration Suite	trial	Dec 23, 2025	Subscribed
SAP Business Application Studio	trial	Dec 23, 2025	Subscribed

Figure 2 : Subscription of SAP Integration Suite

- Next, assign a new role named “Integration Provisioner” so that you’re able to monitor the integration flow. Figure 3 shows that the roles have been assigned.

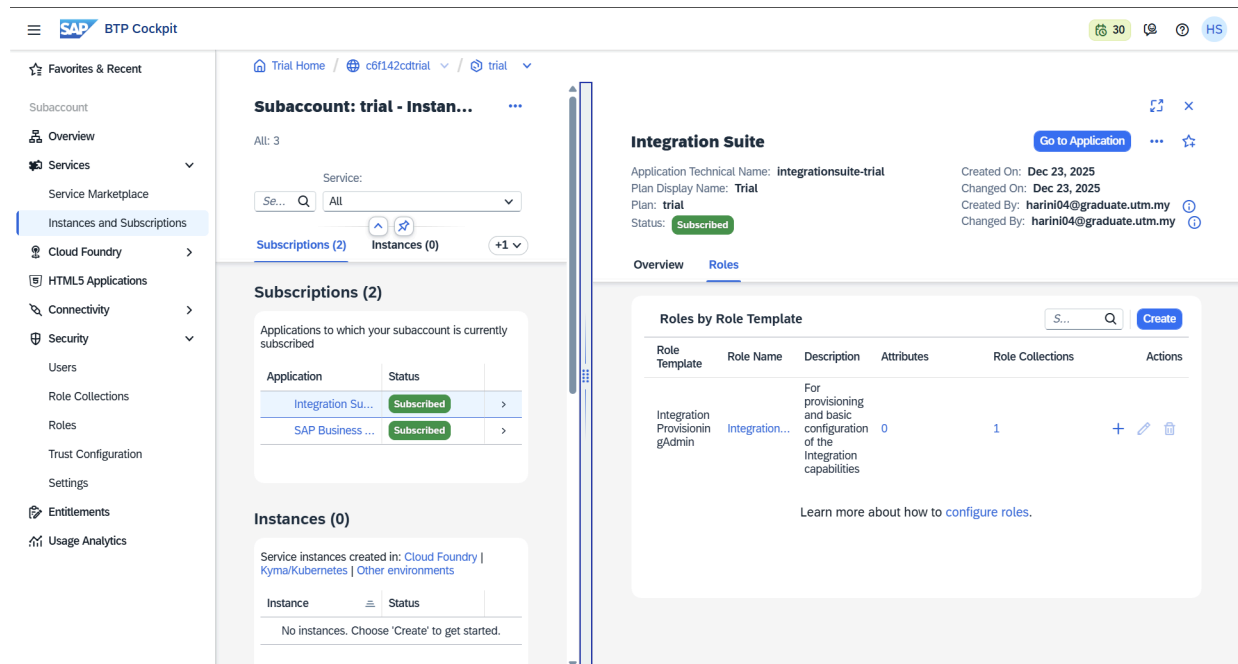


Figure 3 : Role Assignment for Integration Provisioning Admin

- In the SAP Integration Suite page, add capabilities named “Build Integration Scenarios” and “Extend Non-SAP Connectivity” as per Figure 4.

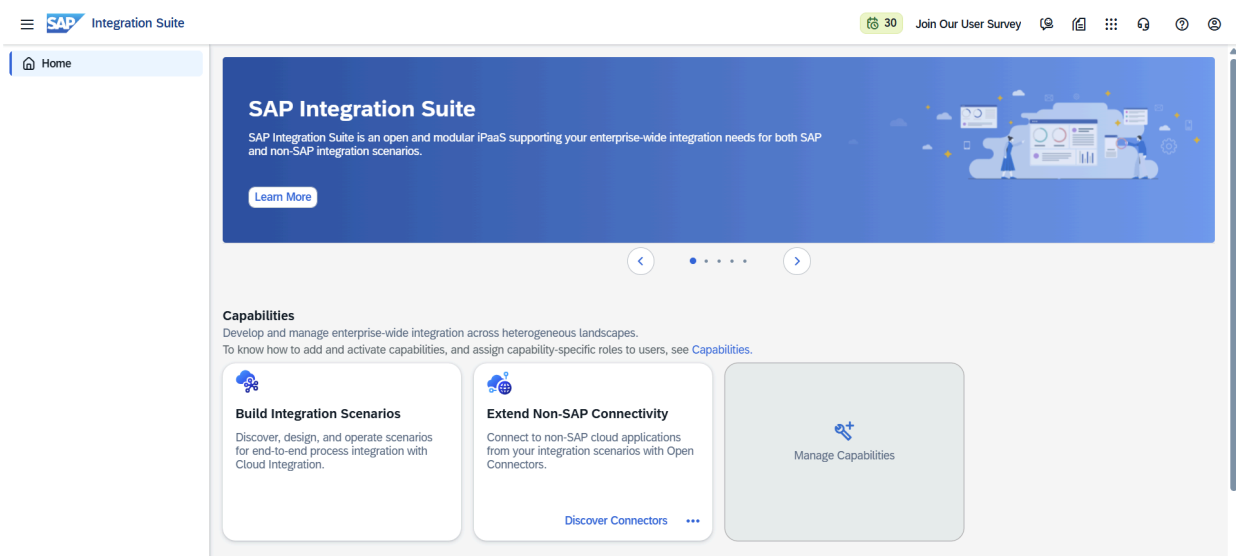


Figure 4 : Adding “Build Integration Scenarios” and “Extend Non-SAP Connectivity” capabilities

4. Add on more roles such as PI_Administrator, PI_Business_Expert, PI_Integration_Developer and OpenConnects_User as per Figure 5.

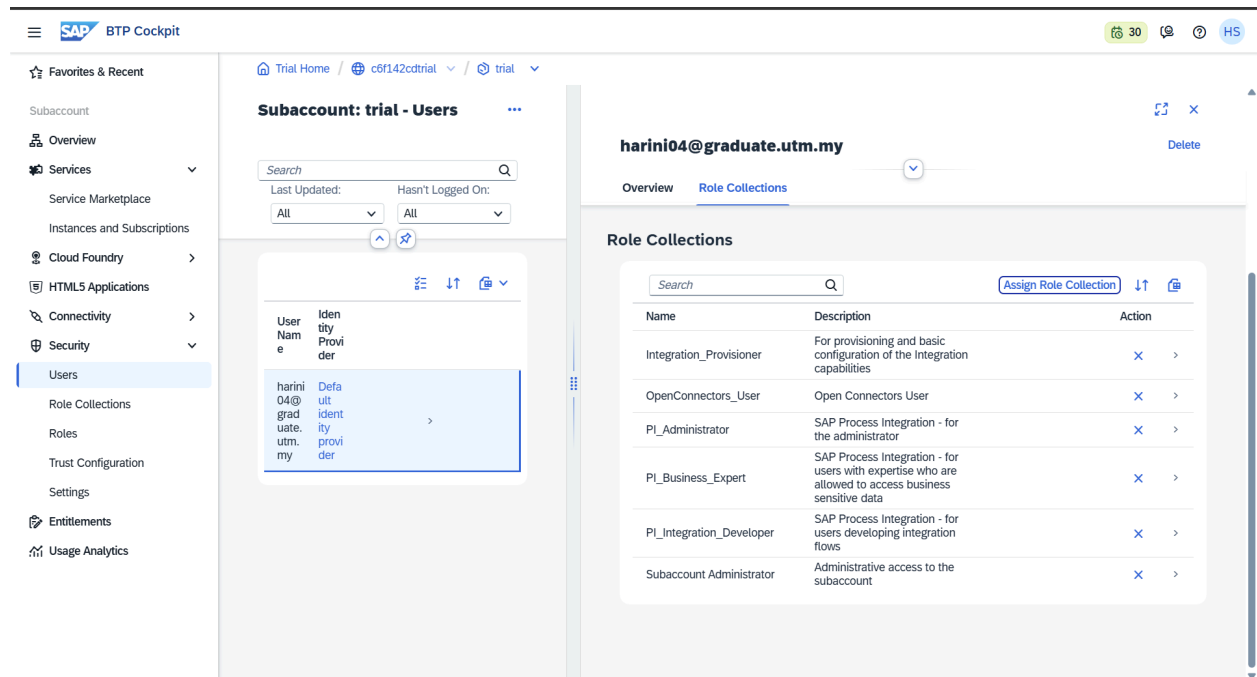


Figure 5 : Role Assignment for PI_Administrator, PI_Business_Expert, PI_Integration_Developer and OpenConnects_User

5. Then, create a service instance and service key. A service instance defines how a service of SAP BTP can be called from a remote component. The service key generated out of the service instance contains credentials and other information that is required for authentication to execute and monitor integration artifacts. Figure 6 shows that a service instance named “process-integration” was created.

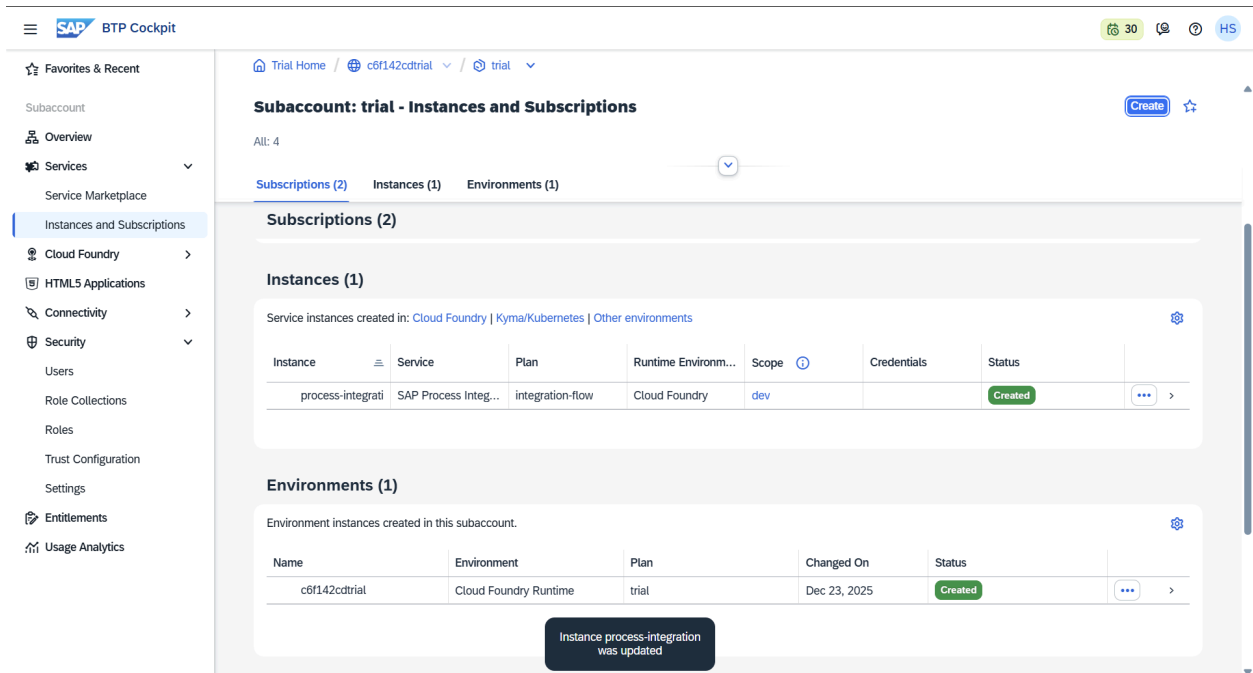


Figure 6 : Successful Service Instance Creation

- The service key is then generated. A clientid and clientsecret is provided. The credentials are as Figure 7.

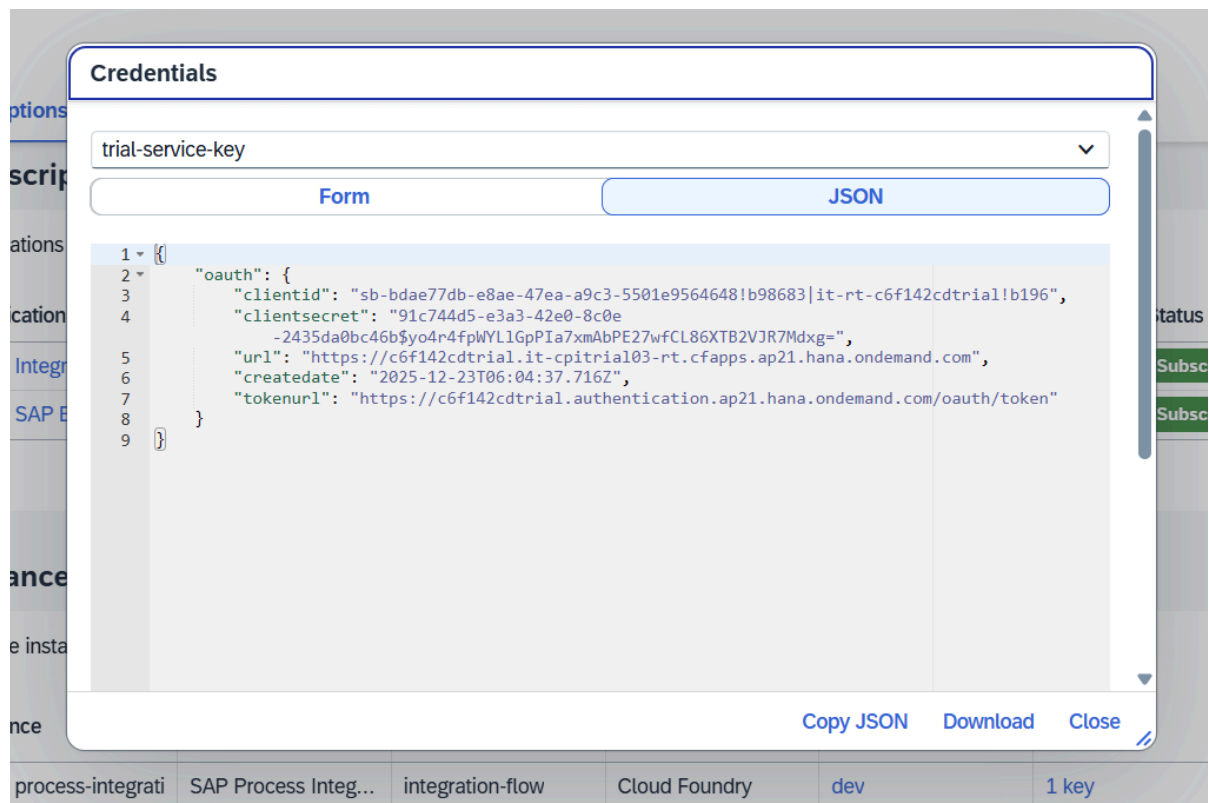


Figure 7 : Credentials of Service Key

7. Sign up for Zoho CRM V2 and load dummy data to access. Figure 8 shows the dummy data accessed.

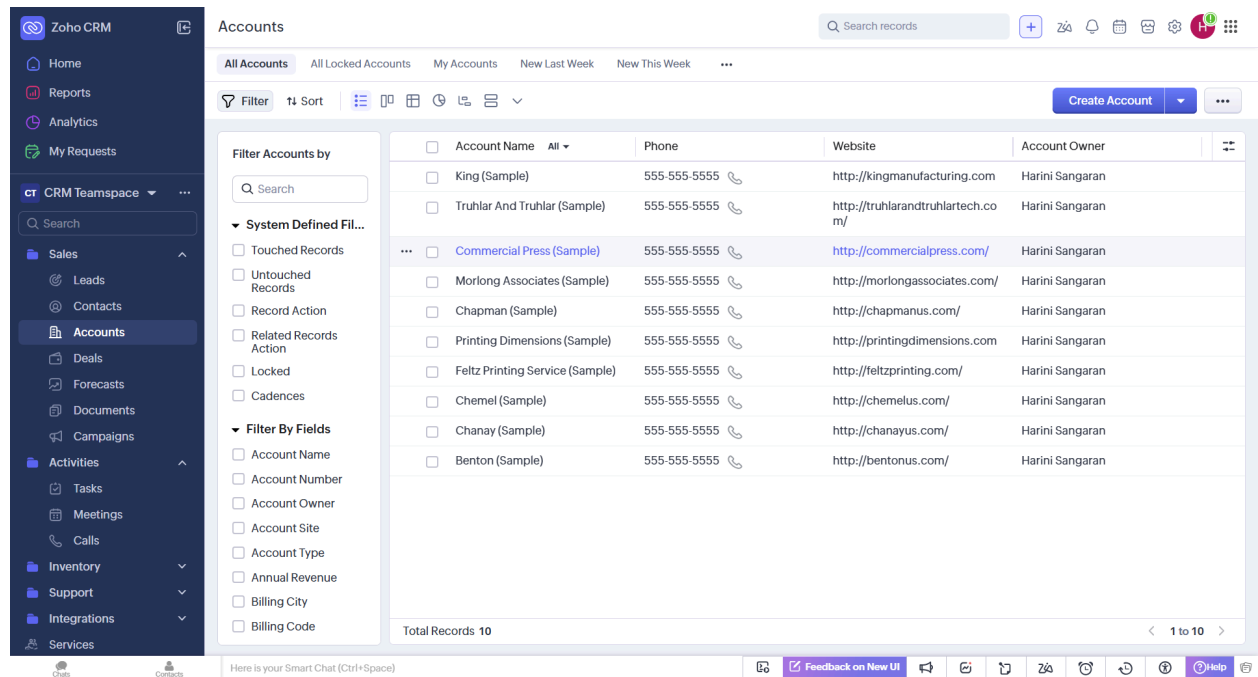


Figure 8 : Dummy Data from Zoho CRM V2

8. Next, generate an API Key. Remember or store the clientid and clientserver given for future use. Figure 9 shows the clientid and clientserver after generating the API Key.

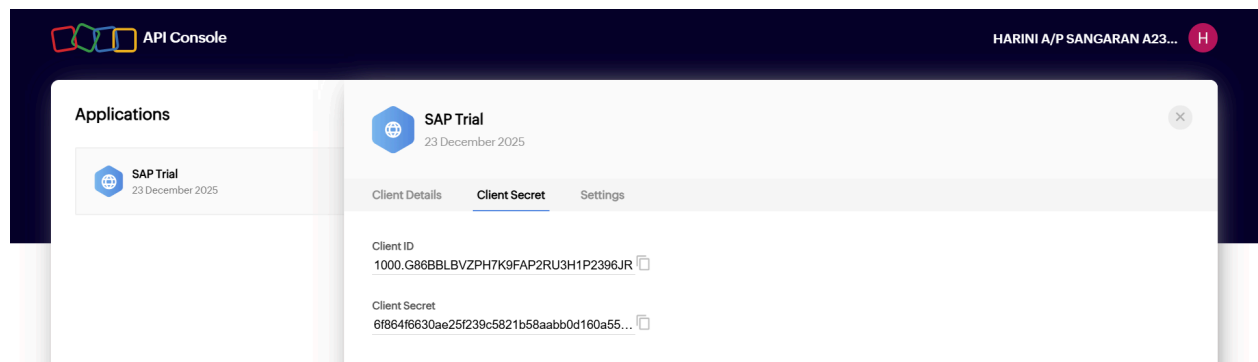


Figure 9 : clientid and clientserver of Zoho CRM V2

9. Go to SAP Integration Suite homepage and click on the “Discover Connectors” button under “Extend Non-SAP Connectivity” capability to see the list of connectors available under SAP. Find and click “API Docs” under Zoho CRM V2 as per Figure 10.

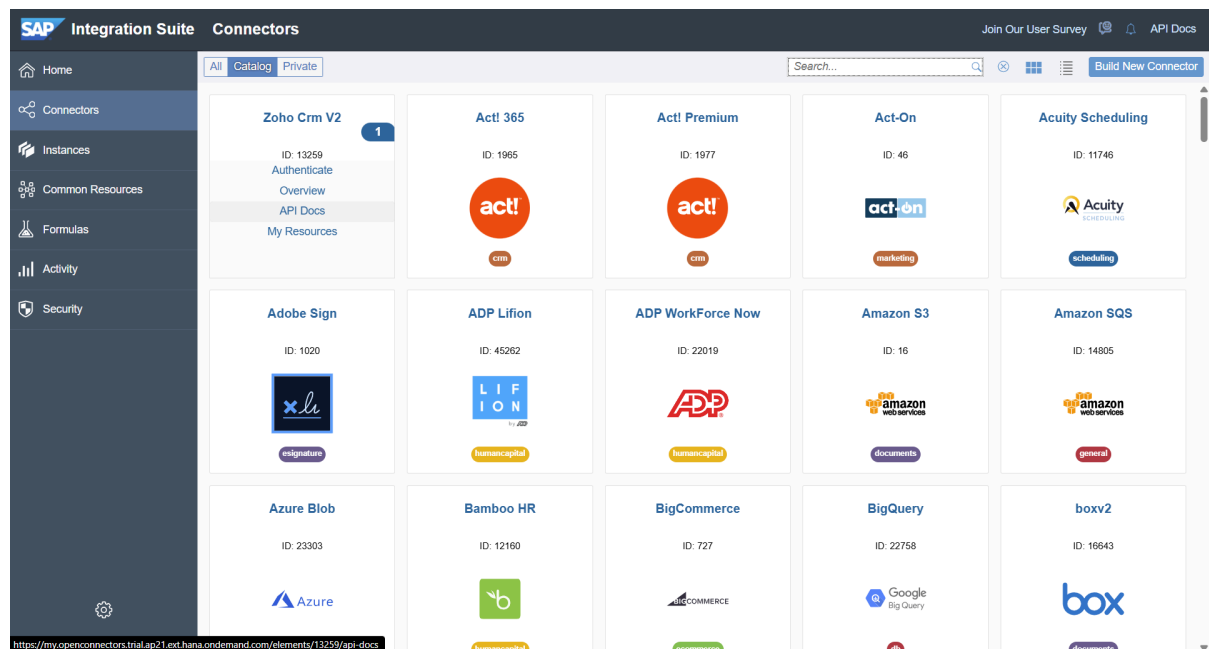


Figure 10 : List of Connectors Under SAP

- Click on the “Authenticate Instance” button. Based on Figure 11, you can see that an instance named “ZohoTutorial_V2” has been created to try out this tutorial.

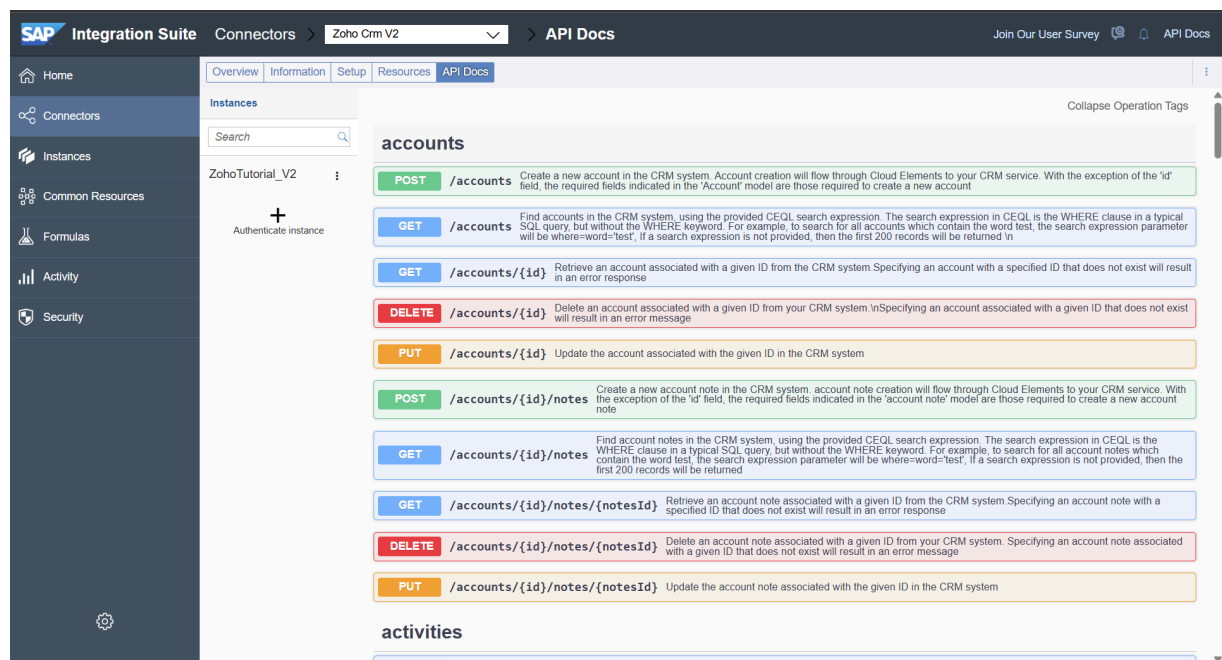


Figure 11 : List of Components of The Dummy Data

- Create an instance by giving it a name. Click on the “Show Optional Fields” button and fill up the OAuth API Key with the clientid of Zoho CRM V2 and the OAuth API Secret with the clientsecret of Zoho CRM V2. Click on the “Create Instance” button to create the new

instance. Figure 12 shows the complete information inserted in each column before creating the instance.

The screenshot displays the 'Create' page for the 'Zoho Crm V2' connector in the SAP Integration Suite. The left sidebar contains navigation links: Home, Connectors, Instances, Common Resources, Formulas, Activity, and Security. The main content area is divided into two panels. The left panel, titled 'Zoho Crm V2', contains the following fields:

- Name:** Tutorial
- Configuration:**
 - OAuth API Key:** 1000.G86BBLBVZPH7K9FAP2RU3H1P2396JR
 - OAuth API Secret:** *****
 - Zoho API Base Uri:** https://www.zohoapis.com
 - Zoho Accounts Base Uri:** https://accounts.zoho.com
- Instance Variables:** A text area containing a JSON object: `{ "name": "Tutorial ", "providerData": { "code": "<INSERT>" } }`

At the bottom of the left panel are buttons for 'Revisions', 'Cancel', and 'Create Instance'. The right panel, titled 'CURL', displays a curl command for a GET request to the Zoho API endpoint, with placeholders for the API key, secret, and OAuth code.

Figure 12 : Information Needed To Create A New Instance

12. To test out if the API is working, click on any row that displays “GET” to retrieve information. In this tutorial, “/accounts” under section accounts is used. Click on the column that you would like to retrieve data from and click on the “Try it out” button. Then, click on the “Execute” button. Figure 13 shows the interface of this process. If your API is working, the results should be as Figure 14. Remember or store the information in the “Curl” and “Request URL” column for future use.

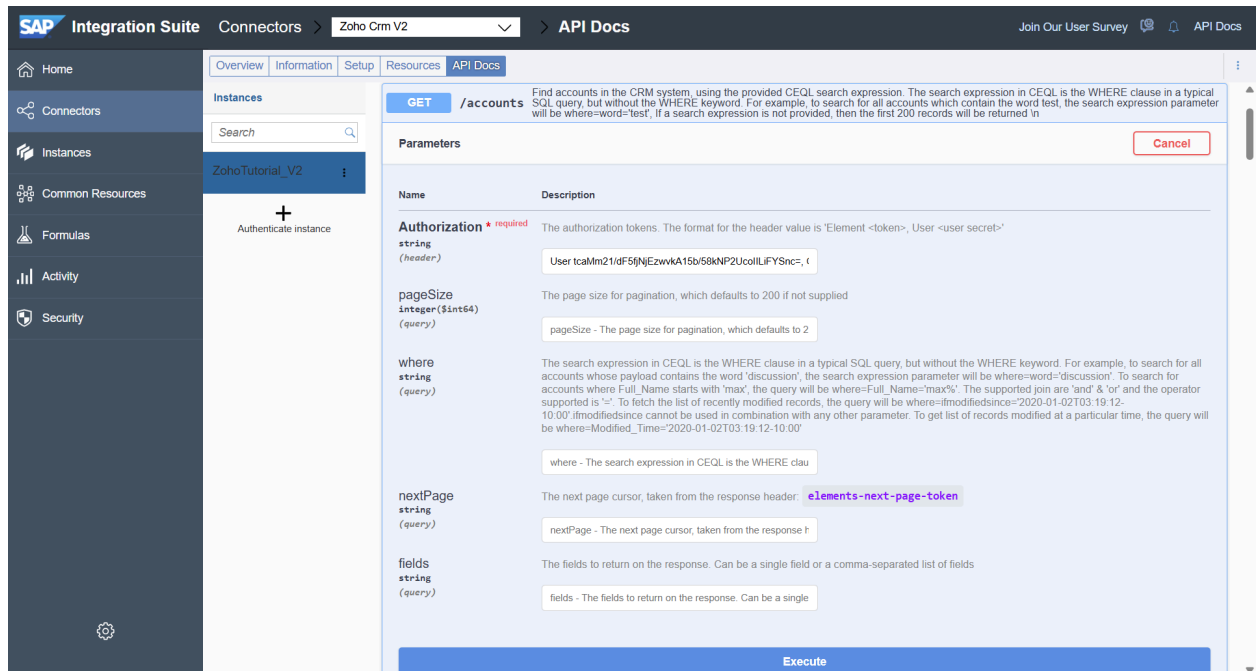


Figure 13 : Interface Before Executing The API

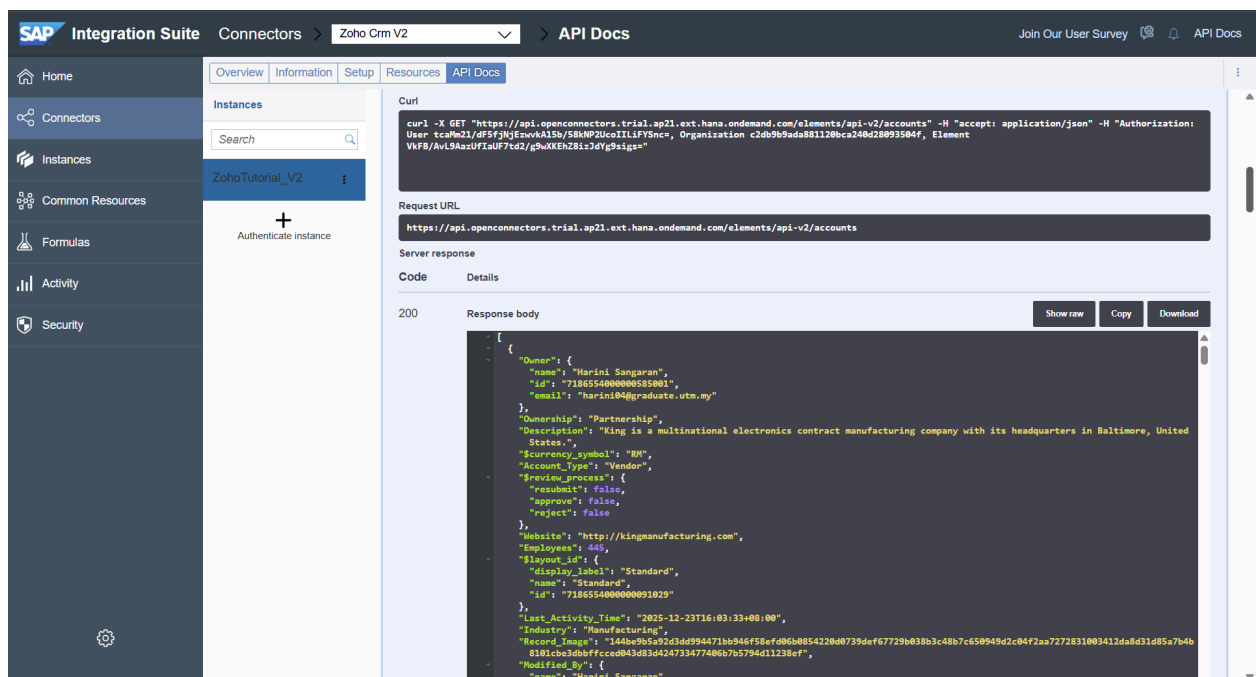


Figure 14 : Successful API Connection

13. An integration between Cloud Integration and Zoho CRM V2 will be established through Open Connectors. For this integration to work as expected, Open Connectors must authenticate via a user credential. The authorization values are provided in the “Curl”

column. Go back to SAP Integration Suite. Under Security Materials, create user credentials based on the values given in the “Curl” column. Figure 15 shows that a user credential has been created.

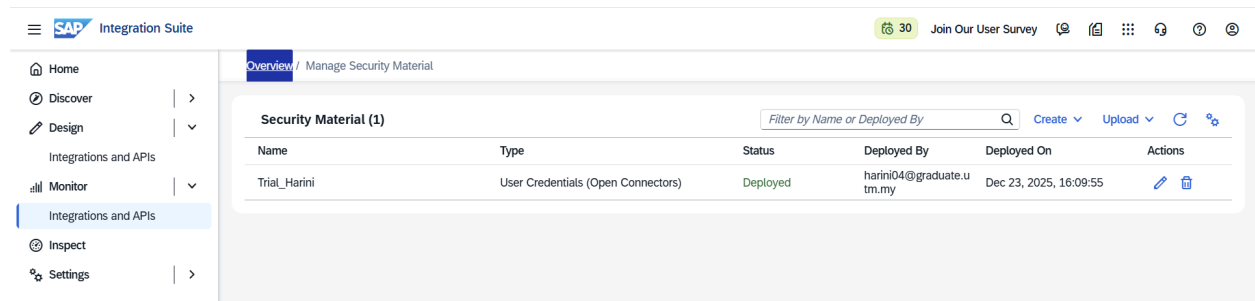


Figure 15 : Successful User Credential Creation

14. In the SAP Integration Suite page, click on “Design” in the toolbar and click “Integration and APIs”. Create an integration package. Then, under the integration package, click on “Artifacts” and create an integration flow. Figure 16 shows an integration package named “Community Challenge” and an integration flow named “Non-SAP_Connectivity” has been created.

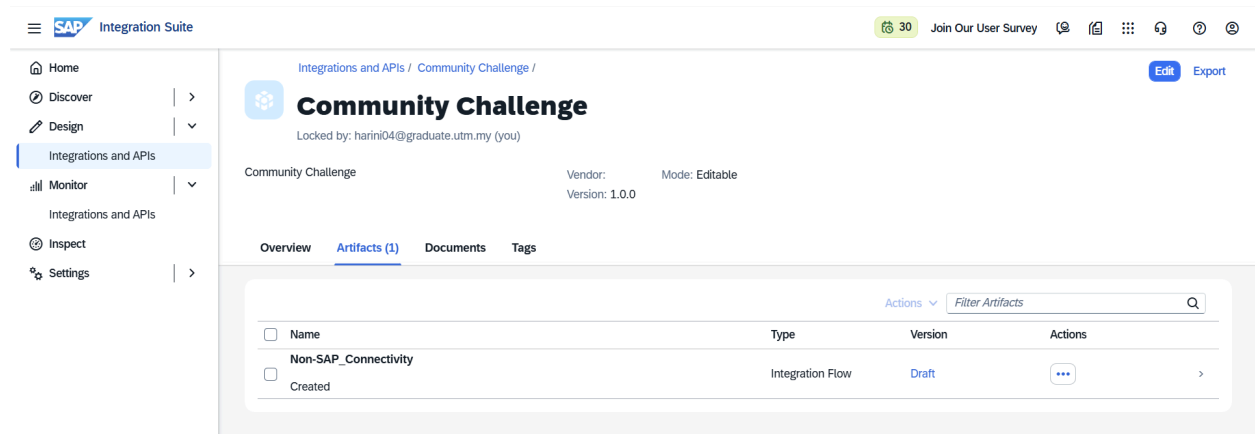


Figure 16 : Successful Integration Package and Integration Flow Creation

15. Once you click on the integration flow, you’ll see an incomplete integration flow as per Figure 17. Drag an arrow from the “Sender” to “Start” and select “HTTPS” as the adapter type. Under HTTPS’ “Connection” tab, enter “/accounts/getdetails” as the address. Then, click on the arrow between “Start” and “End” and select “External Call” under the “Call” button and select “Request Reply”. Next, drag an arrow from “Request Reply” to the “Receiver”. Select “OpenConnectors” as the adapter type. Click on the “Connection” tab

under “OpenConnectors” and fill the Base URL from the successful API run gathered in Figure 14 but up till **/api-v2** only. For example, If your request URL is `https://<request-URL>/elements/api-v2/employees`, paste only `https://<request-URL>/elements/api-v2`. Next, fill up the resource column as “/accounts” for our use case and set the method as “GET”. Save and deploy the integration flow. The complete integration flow should look like Figure 18.

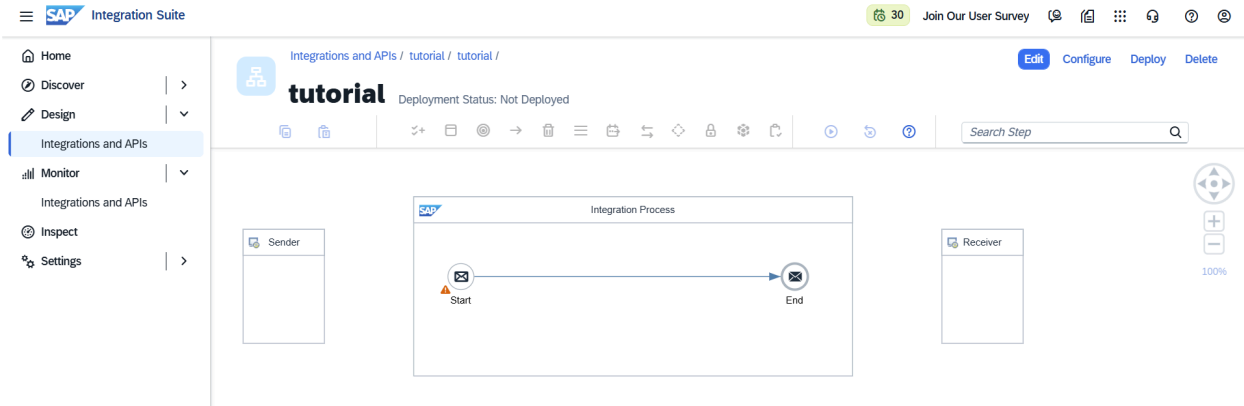


Figure 17 : Incomplete Integration Flow

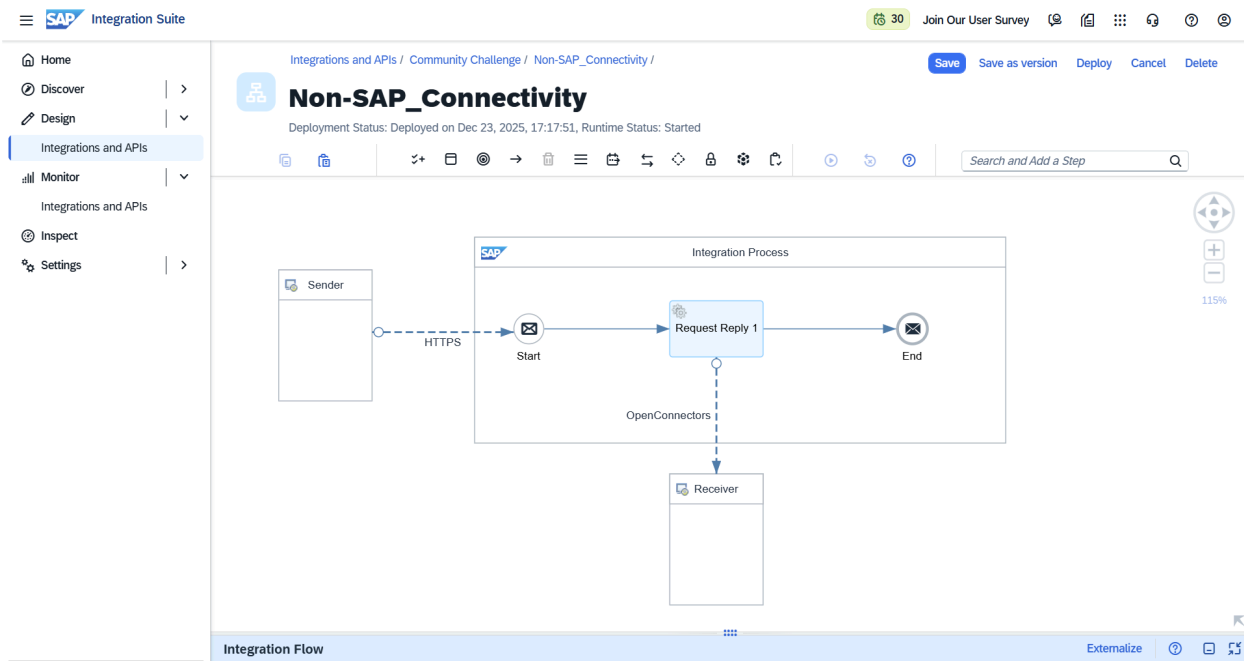


Figure 18 : Complete Integration Flow

16. Go back to the SAP Integration Suite and click on “Monitor”. Then, click on “Integration and APIs”. Click on “started” under Manage Integration Content. Copy the endpoints URL. Figure 19 shows the interface.

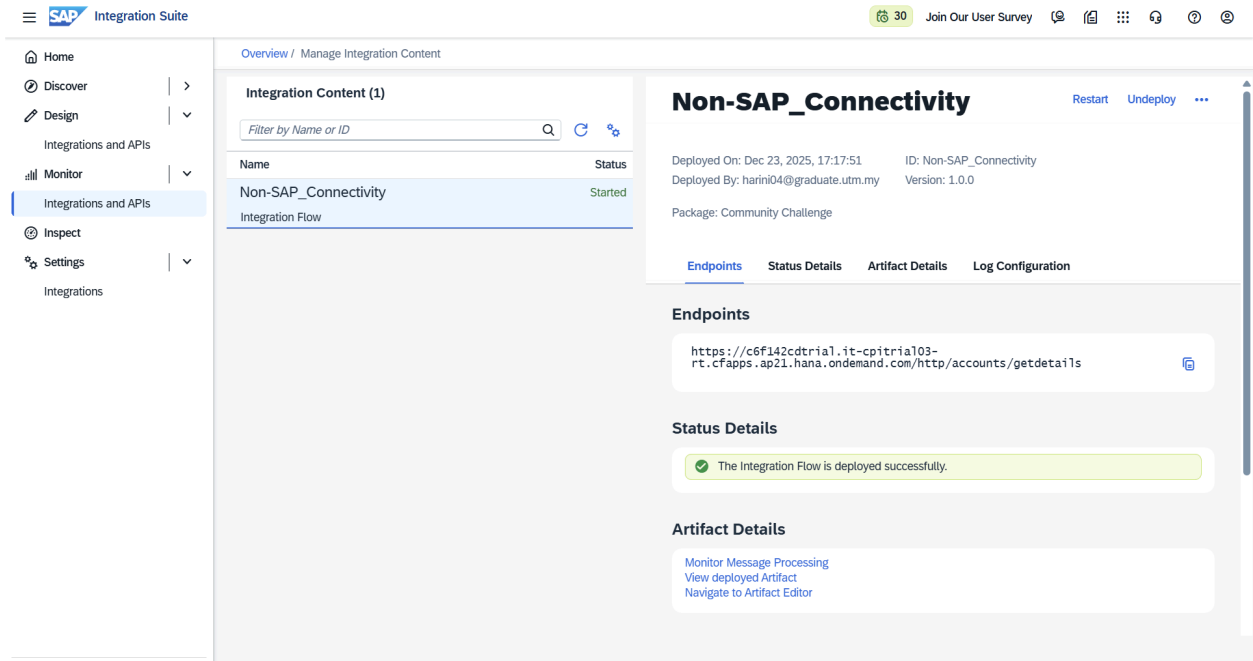


Figure 19 : Interface with Endpoints URL

17. Now, to test out the integration scenario, go to the Postman application. Paste the endpoint URL in the URL box and set the method as GET. Click on the dropdown showing “Params” and select “Authorization”. Then, select “Basic Auth”. Enter clientid of service key as username and the clientsecret of service key as the password. The credentials can be found in Figure 7. Then, click “Send” button. If the API runs successfully, the body should display information fetched as shown in Figure 20.

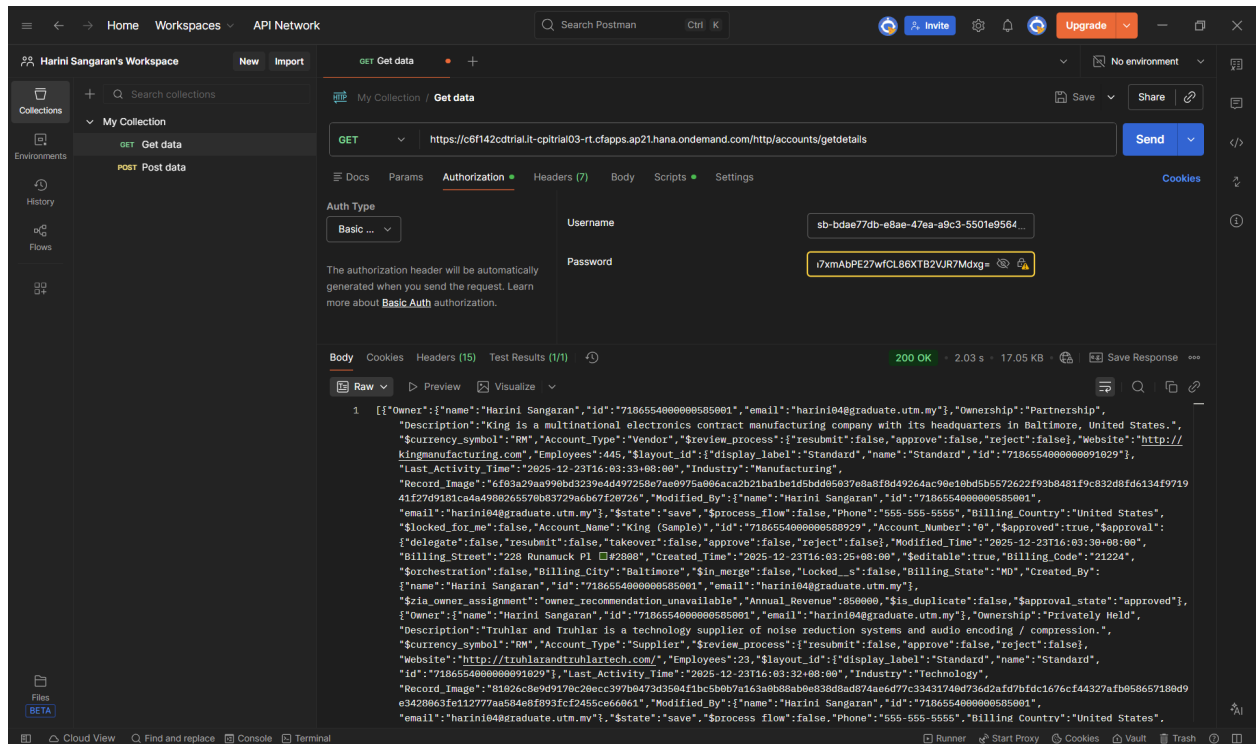


Figure 20 : Information Fetched Via API Being Displayed in Postman Application

Conclusion

This project demonstrates how SAP Open Connectors offers a standardized interface for a variety of data sources, thereby reducing the complexity of third-party APIs. The usage of Cloud Integration shows that a fully managed integration flow can be deployed and monitored in a cloud-based runtime environment. The technical challenges encountered, specifically being unable to create a BambooHR account served as critical learning points. I discovered more and gained a better understanding when exploring Zoho CRM V2. I underscored the necessity of precise configuration in enterprise-grade integrations, particularly the importance of matching security permissions (scopes) and ensuring regional alignment between the SAP Data Center and the third-party provider. Ultimately, the successful retrieval of Zoho "accounts" data via Postman validates that SAP BTP provides a robust, scalable framework for real-world business process automation.